

Introduction to the Course

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Financial Data Science

University of Pavia

29th September 2025

Outline

- 1 General Information
 - Overview of the course
 - Schedule of the Financial Data Science course

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Structure of the Financial Data Science course

The course is split into two modules, one composed of 23 and one composed of 22 hours.

Specifically, the two modules are held by

- a) **Emanuela Raffinetti** (23h)
 - b) **Lorenzo Trapani** (22h)
- The course is addressed to the presentation of the main Machine and Deep Learning models, factor models, estimation techniques, together with innovative approaches for ensuring the trustworthiness condition of Artificial Intelligence systems.
 - **Assessment methods and criteria** - The examination procedure is based on an oral slide-presentation and discussion of the main data analysis results.

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Planning

FINANCIAL DATA SCIENCE A.Y. 2025/2026

Schedule

29th September 2025 (4h) Emanuela Raffinetti	• Linear regression (theory) • Application in R
6th October 2025 (4h) Emanuela Raffinetti	• Logistic regression (theory) • Application in R
13th October 2025 (4h) Emanuela Raffinetti	• Model selection (theory) • Application in R
20th October 2025 (4h) Emanuela Raffinetti	• Tree and Random Forest models (theory) • Application in R
27th October 2025 (4h) Emanuela Raffinetti	• Neural networks models (theory) • Application in R
3rd November 2025 (4h) Lorenzo Trapani	• Further linear regression estimators – part I • Application in R
10th November 2025 (4h) Lorenzo Trapani	• Further linear regression estimators – part II • Application in R
17th November 2025 (4h) Lorenzo Trapani	• Large dimensional factor models – representation and properties • Large dimensional factor models – estimators of the number of common factors
24th November 2025 (4h) Lorenzo Trapani	• Large dimensional factor models – estimators of factors and loadings • Applications in R
1st December 2025 (4h) Lorenzo Trapani	• Large dimensional factor models – asset pricing and the factor zoo – part I • Applications in R
15th December 2025 (3h) Emanuela Raffinetti	• S.A.F.E. AI (Sustainable [Robust], Accurate, Fair, Explainable AI) • Application in R
12th January 2026 (2h) Lorenzo Trapani	• Large dimensional factor models – asset pricing and the factor zoo – part II • Applications in R

Bibliography

Possible bibliographic references are:

- Lantz B. (2013): Machine Learning with R, PUCKY Publishing open source
- Gareth J., Witten D., Hastie T., Tibshirani R. (2023): An Introduction to Statistical Learning, 2nd Edition, Springer, downloadable from the website: <https://www.statlearning.com>
- Giudici P. (2003): Applied Data Mining - Statistical Methods for Business and Industry, 1st Edition, Wiley, available at: https://theswissbay.ch/pdf/Gentoomen%20Library/Data%20Mining/Applied%20Data%20Mining-Statistical%20Methods%20for%20Business%20and%20Industry_Giudici%20P%20%282003%29.pdf
- Scientific papers and additional books will be also mentioned during the course